Conditional Expectation-Covariance

Definition (Independence two Discrete r.v.s)

Two Random variables X and Y are independent if for all x and y values

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

Definition (Independence two continuous r.v.s)

Two Random variables X and Y are independent if for all x and y values

$$f(X = x, Y = y) = f(X = x)f(Y = y)$$

Eg 6. Two continuous random variables X and Y have joint probability density function

$$f(x, y) = \begin{cases} \dfrac{1}{2xy} & 1 \leq x \leq e, 1 \leq y \leq e^2 \\ 0 & \text{otherwise} \end{cases}$$

Are X and Y independent?

In example (3-a) on page 8, we showed that

$$f(x) = \begin{cases} \dfrac{1}{x} & 1 \leq x \leq e \\ 0 & \text{otherwise} \end{cases} \qquad f(y) = \begin{cases} \dfrac{1}{2y} & 1 \leq y \leq e^2 \\ 0 & \text{otherwise} \end{cases}$$

for any x and any y we have

$$f(X=x, Y=y) = f_X(x) \cdot f_Y(y)$$

Hence, X and Y are independent.

Definition (Conditional PMF)

For discrete r.v.s X and Y, the conditional PMF of Y given $X = x$ is

$$P(Y = y|X = x) = \frac{P(X = x, Y = y)}{P(X = x)}$$

Eg 1. Let X and Y be two random variable which are jointly distributed with pmf $p_{X,Y}(x, y)$ as follows

$y = 1$	$\frac{5}{12}$	$\frac{1}{4}$
$y = 0$	$\frac{1}{4}$	$\frac{1}{12}$
	$x = 0$	$x = 1$

Calculate $Var(X|Y = 0)$.

$X | Y=0 \sim \text{Bern}(p)$

where $p = P(X=1 | Y=0)$

$$p = P(X=1|Y=0) = \frac{P(X=1, Y=0)}{P(Y=0)}$$

$$p = \frac{\frac{1}{12}}{\frac{1}{4}+\frac{1}{12}} = \frac{\frac{1}{12}}{\frac{4}{12}} = \frac{1}{4}$$

Since $X \mid Y=0 \sim \text{Bern}(p=\frac{1}{4})$, hence

$$\text{var}(X \mid Y=0) = p(1-p) = \frac{1}{4}\left(1-\frac{1}{4}\right) = \frac{1}{4}\cdot\frac{3}{4}$$

$$\Rightarrow \text{var}(X \mid Y=0) = \frac{3}{16}.$$

Second method:

$$\text{var}(X \mid Y=0) = E[X^2 \mid Y=0] - \left(E[X \mid Y=0]\right)^2$$

$$E[X^2 \mid Y=0] = \sum_x x^2 \cdot P(X=x \mid Y=0)$$

$$= 0 \cdot P(X=0 \mid Y=0) + 1^2 \cdot P(X=1 \mid Y=0) = \frac{1}{4}$$

$$E[X \mid Y=0] = \sum_x x \, P(X=x \mid Y=0)$$

$$= 0 \cdot P(X=0 \mid Y=0) + 1 \cdot P(X=1 \mid Y=0) = \frac{1}{4}$$

$$\text{var}(X \mid Y=0) = \frac{1}{4} - \left(\frac{1}{4}\right)^2 = \frac{1}{4} - \frac{1}{16} = \frac{3}{16}.$$

Definition (Conditional PDF)

For continuous r.v.s X and Y, the conditional PDF of Y given $X = x$ is

$$f(Y = y|X = x) = \frac{f(X = x, Y = y)}{f(X = x)} = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

Eg 2-A. The joint density function of X and Y is given by

$$f(x, y) = \begin{cases} \dfrac{x + y}{3} & 0 < x < 1,\ 1 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

Compute the conditional density of X, given that $Y = \dfrac{3}{2}$.

we should find $f_{X|Y}(x|\frac{3}{2}) = \dfrac{f(X=x, Y=\frac{3}{2})}{f_Y(\frac{3}{2})}$

$$f_{X|Y}(x|\tfrac{3}{2}) = \frac{f(X=x, Y=\frac{3}{2})}{f_Y(\frac{3}{2})} = \begin{cases} \dfrac{\frac{x+\frac{3}{2}}{3}}{f_Y(\frac{3}{2})} & \text{if } 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

we should find $f_Y(\frac{3}{2})$, for that first we should find marginal pdf for Y.

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx = \begin{cases} \int_0^1 \frac{x+y}{3}\,dx & 1<y<2 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_0^1 \frac{x+y}{3}\,dx = \frac{1}{3}\int_0^1 (x+y)\,dx = \frac{1}{3}\left(\frac{x^2}{2}+yx\right)\Big|_0^1 = \frac{1}{3}\left(\frac{1}{2}+y\right)$$

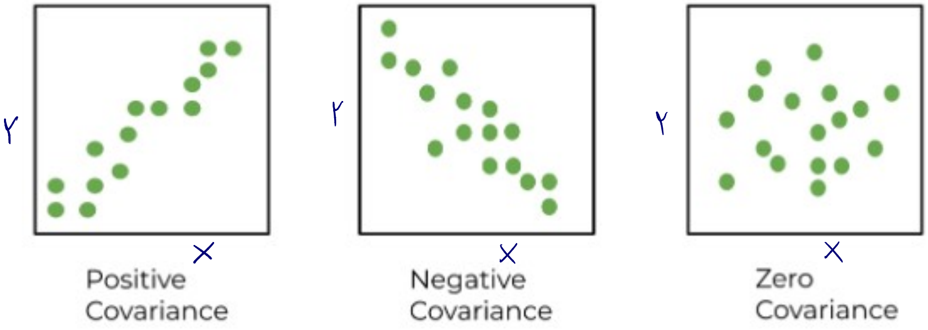
$$\Rightarrow \quad f_Y(y) = \begin{cases} \frac{1}{3}\left(y+\frac{1}{2}\right) & 1<y<2 \\ 0 & \text{otherwise} \end{cases}$$

$$f_Y\left(\tfrac{3}{2}\right) = \frac{1}{3}\left(\frac{3}{2}+\frac{1}{2}\right) = \frac{2}{3}$$

$$f_{X|Y}\left(x \middle| \tfrac{3}{2}\right) = \frac{f\left(X=x, Y=\frac{3}{2}\right)}{f_Y\left(\frac{3}{2}\right)} = \begin{cases} \dfrac{\frac{x+\frac{3}{2}}{\cancel{3}}}{\frac{2}{\cancel{3}}} = \dfrac{\frac{2x+3}{2}}{2} = \dfrac{2x+3}{4} & \text{if } 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

$$\Rightarrow \quad \boxed{f_{X|Y}\left(x \middle| \tfrac{3}{2}\right) = \begin{cases} \frac{2x+3}{4} & \text{if } 0<x<1 \\ 0 & \text{otherwise} \end{cases}}$$

Intuitively: Covariance of two random variables X and Y is a quantity that measures the relationship between X and Y.Its definition is very analogues to variance of a random variable.



$Cov(X,Y) > 0$ $\qquad$ $Cov(X,Y) < 0$ $\qquad$ $Cov(X,Y) = 0$

Definition (Covariance)

Let X and Y be two random variables, then the Covariance of X and Y is

$$
\begin{aligned}
& Cov(X, Y) = E[(X - E[X])(Y - E[Y])] \\
\text{or} \quad & Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] \\
\text{or} \quad & Cov(X, Y) = E[XY] - E[X]E[Y]
\end{aligned}
$$

Eg2-B. The joint density function of X and Y is given by

$$
f(x, y) = \begin{cases} \dfrac{x+y}{3} & 0 < x < 1,\ 1 < y < 2 \\ 0 & \text{otherwise} \end{cases}
$$

Calculate $Cov(X, Y)$.

$$Cov(X,Y) = E[XY] - E[X]\,E[Y]$$

To find $E[X]$ and $E[Y]$ first we should find marginal pdf X and Y. we found $f_Y(y)$ in Example 2.A page 9.

$$f_Y(y) = \begin{cases} \frac{1}{3}\left(y+\frac{1}{2}\right) & 1<y<2 \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) = \begin{cases} \int_1^2 \frac{x+y}{3}\,dy & 0<x<1 \\ 0 & \text{otherwise} \end{cases}$$

$$\int_1^2 \frac{x+y}{3}\,dy = \frac{1}{3}\left[xy+\frac{y^2}{2}\right]_1^2 = \frac{1}{3}\left[2x+2-x-\frac{1}{2}\right]$$

$$= \frac{1}{3}\left[x+\frac{3}{2}\right]$$

$$f_X(x) = \begin{cases} \frac{1}{3}\left(x+\frac{3}{2}\right) & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$E[X] = \int_0^1 x \; f_X(x)\, dx$$

$$E[X] = \frac{1}{3}\int_0^1 \left(x^2 + \frac{3}{2}x\right) dx$$

$$E[X] = \frac{1}{3}\left(\frac{x^3}{3} + \frac{3}{4}x^2\right)\Big|_0^1$$

$$E[X] = \frac{1}{3}\left(\frac{1}{3} + \frac{3}{4}\right) = \frac{13}{36}$$

$$f_Y(y) = \begin{cases} \frac{1}{3}\left(y+\frac{1}{2}\right) & 1 < y < 2 \\ 0 & \text{otherwise} \end{cases}$$

$$E[Y] = \int_1^2 y \, f_Y(y)\, dy$$

$$E[Y] = \frac{1}{3}\int_1^2 \left(y^2 + \frac{y}{2}\right) dy$$

$$E[Y] = \frac{1}{3}\left(\frac{y^3}{3} + \frac{y^2}{4}\right)\Big|_1^2$$

$$E[Y] = \frac{1}{3}\left[\frac{8}{3} + 1 - \frac{1}{3} - \frac{1}{4}\right]$$

$$E[Y] = \frac{1}{3}\left[\frac{7}{3} + \frac{3}{4}\right] = \frac{37}{36}$$

$$E[XY] = \int_1^2 \int_0^1 xy \, \frac{x+y}{3} \, dx \, dy$$

$$E[XY] = \int_1^2 \frac{y}{3} \int_0^1 (x^2 + xy) \, dx \, dy$$

$$E[XY] = \int_1^2 \frac{y}{3} \left(\frac{x^3}{3} + \frac{x^2 y}{2} \right) \Big|_0^1 \, dy$$

$$E[XY] = \int_1^2 \frac{y}{3} \left(\frac{1}{3} + \frac{y}{2} \right) dy = \frac{1}{18} \int_1^2 \overbrace{y(2+3y)}^{2y+3y^2} \, dy$$

$$= \frac{1}{18} \left[y^2 + y^3 \right]_1^2 = \frac{1}{18} [4 + 8 - 2] = \frac{10}{18}$$

$$\mathrm{Cov}(X,Y) = E[XY] - E[X]E[Y] = \frac{10}{18} - \frac{13}{36} \cdot \frac{37}{36} = \frac{1}{18}\left[10 - \frac{13}{2} \, \frac{37}{36} \right]$$

$$\mathrm{Cov}(X,Y) = \frac{1}{18} \left[\frac{720 - 481}{72} \right] = \frac{239}{1296}$$

You should know the followings by heart.

Properties of Covariance

(1) $Cov(X, X) = Var(x)$

(2) $Cov(X, Y) = Cov(Y, X)$

(3) $Cov(X, c) = 0$, where c is a constant number.

(4) $Cov(cX, Y) = cCov(X, Y)$ where c is a constant number.

(5) $Cov(X, Y + Z) = Cov(X, Y) + Cov(X, Z)$

(6) $Cov(\sum_{i=1}^{n} X_i, \sum_{i=1}^{n} X_j) = \sum_{i=1}^{n} Var(X_i) + 2 \sum \sum_{i<j} Cov(X_i, X_j)$

Eg. Show that Var(X+Y)= Var(X) + Var(Y) + 2 Cov(X,Y)

$$\mathrm{var}(X+Y) = \mathrm{cov}(X+Y, X+Y) = \mathrm{cov}(X,X) + \mathrm{cov}(X,Y) + \mathrm{cov}(Y,X)$$

$$+ \mathrm{cov}(Y,Y) = \mathrm{cov}(X,X) + 2\,\mathrm{cov}(X,Y) + \mathrm{cov}(Y,Y)$$

$$= \mathrm{var}(X) + 2\,\mathrm{cov}(X,Y) + \mathrm{var}(Y)$$